

DIRECTIONS for questions 1 to 5: Answer the questions on the basis of the information given below.

Lokesh, who trades in the commodity market, was interested in investing in two commodities – Commodity A and Commodity B. He followed the price of these two commodities for n days, from Day 1 to Day n , and he observed that they followed the following trends during the given period:

- At the beginning of a day, if the price of Commodity A was greater than that of Commodity B, the price of Commodity A increased by ₹10 and the price of Commodity B increased by ₹40 by the beginning of the next day.
- At the beginning of a day, if the price of Commodity A was less than or equal to that of Commodity B, the price of Commodity A increased by ₹30 and the price of Commodity B increased by ₹10 by the beginning of the next day.

At the beginning of Day 1, the price of Commodity A was ₹130 and the price of Commodity B was ₹140.

Consider that the price of a commodity indicates the price only at the beginning of a day.

Q1. DIRECTIONS for question 1: Type in your answer in the input box provided below the question.

If $n = 15$, on how many days was the price of Commodity A greater than that of Commodity B?

Q2. DIRECTIONS *for question 2:* Select the correct alternative from the given choices.

If $n = 100$, what was the maximum difference between the prices of Commodity A and Commodity B on any day?

- ☐ a) ₹740
- ☐ b) ₹1240
- ☐ c) ₹20
- ☐ d) None of the above

Q3. DIRECTIONS *for question 3:* Type in your answer in the input box provided below the question.

If Lokesh observed that the price of at least one of the two commodities was less than ₹500 on the n^{th} day, what is the maximum possible value of n ?

Q4. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.

If $n = 73$, what is the difference between the average price (in ₹) of Commodity A and that of Commodity B?

- ☐ a) 0.256
- ☐ b) 0.274
- ☐ c) 0.265
- ☐ d) 0.233

Q5. DIRECTIONS *for questions 4 and 5:* Select the correct alternative from the given choices.

On which of the following days was the price of Commodity B be greater than that of Commodity A?

- ☐ a) **Day 77**
- ☐ b) **Day 35**
- ☐ c) **Day 93**
- ☐ d) **Day 104**

DIRECTIONS *for questions 6 to 10:* Answer these questions on the basis of the information given below.

Himesh, the manager of a gym, has six different weighing machines – Machine A through Machine F – in his gym. However, except for one, none of the machines display the correct weight. On a particular day, Himesh measured his weight on each of the six machines and each machine displayed a different weight among 50 kg, 53 kg, 58 kg, 61 kg, 64 kg and 70 kg, in no particular order.

The following information is known about the weights displayed by the machines:

- i. The weight displayed by Machine A was more than that displayed by Machine D but neither of them displayed the correct weight.
- ii. The weight displayed by Machine F was 3 kg more than the correct weight of Himesh.
- iii. The machine that displayed the correct weight did not record the lowest weight among the six machines.
- iv. The weight displayed by Machine C was 6 kg more than that displayed by Machine B, which did not record the correct weight.

Q6. DIRECTIONS *for question 6:* Type in your answer in the input box provided below the question.

What is the correct weight (in kg) of Himesh?

Q7. DIRECTIONS *for question 7:* Select the correct alternative from the given choices.

What is the difference (in kg) between the weight displayed by Machine D and that displayed by Machine C?

☐ a) 14

☐ b) 3

☐ c) 20

☐ d) 12

Q8. DIRECTIONS *for question 8: Type in your answer in the input box provided below the question.*

What was the weight (in kg) displayed by Machine A?

Q9. DIRECTIONS *for questions 9 and 10:* Select the correct alternative from the given choices.

If the error (in kg) in measurement of each machine is constant, what will be the weight displayed (in kg) by Machine B for a dumbbell weighing 20 kg?

☐ a) 20

☐ b) 31

☐ c) 23

☐ d) 26

Q10. DIRECTIONS *for questions 9 and 10:* Select the correct alternative from the given choices.
How many machines displayed a weight lower than the correct weight of Himesh?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) 5

DIRECTIONS *for questions 11 to 15:* Answer the questions on the basis of the information given below.

In a class, six students, A through F, were sitting in six chairs from left to right, all facing the same direction, not necessarily in the same order. Karthik, the teacher of the class, handed over a certain number of blank sheets to the student at the extreme left. Karthik asked each student to take as many blank sheets as he wants and pass the remaining to the student to his immediate right. After the sixth student (i.e., the student sitting at the extreme right) took the sheets, Karthik collected the remaining sheets from him.

It is also known that

- i. the number of sheets that the first student took was twice the number of sheets that C took.
- ii. no two students took the same number of sheets and each student took at least one sheet.
- iii. B, who was not sitting at the extreme right, took two sheets more than the student who was sitting two places to his left.
- iv. D, who was sitting to the immediate left of B, took 15 sheets and the number of sheets that any other student took was either greater than 20 or less than 10.
- v. E, who was sitting two places to the right of the student sitting at the extreme left, took 6 sheets.
- vi. Karthik gave 60 sheets to the student sitting at the extreme left and collected 18 sheets from F.

Q11. DIRECTIONS *for question 11:* Type in your answer in the input box provided below the question.

How many sheets did C take?

Q12. DIRECTIONS *for question 12:* Select the correct alternative from the given choices.
Who was sitting to the immediate left of the person who took the lowest number of sheets?

- ☐ a) **A**
- ☐ b) **E**
- ☐ c) **D**
- ☐ d) **C**

Q13. DIRECTIONS *for question 13:* Type in your answer in the input box provided below the question.

What is the difference between the number of sheets that A and D took?

Q14. DIRECTIONS *for questions 14 and 15:* Select the correct alternative from the given choices.
Which of the following statements is definitely true?

- ☐ a) The student sitting at the extreme right took the highest number of sheets.
- ☐ b) The student sitting at the extreme left took the highest number of sheets.
- ☐ c) The student sitting to the immediate right of **B** took the highest number of sheets.
- ☐ d) The student sitting to the immediate left of **E** took the lowest number of sheets.

Q15. DIRECTIONS *for questions 14 and 15:* Select the correct alternative from the given choices.

How many students took more than 5 sheets?

☐ a) 6

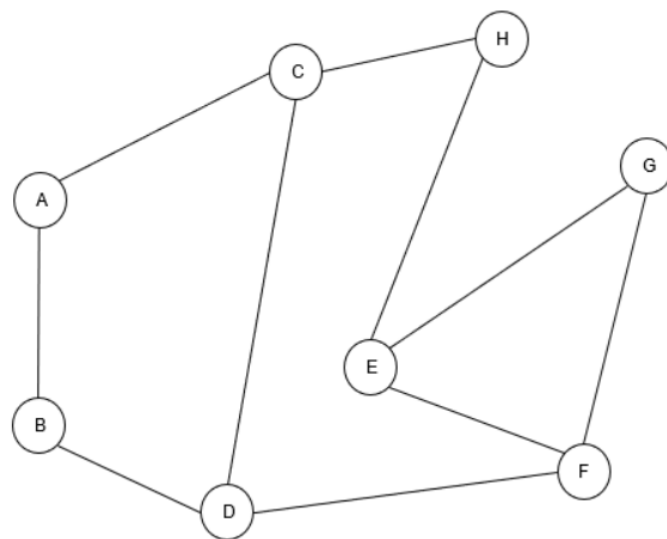
☐ b) 5

☐ c) 4

☐ d) 3

DIRECTIONS for questions 16 to 20: Answer the questions on the basis of the information given below.

Eight cities, A through H, are connected by a network of two-way roads as shown in the figure below:



Kiran, a travelling salesman, has to travel from one of the eight cities to another, such that he visits each of the eight cities exactly once. The city from which he starts is referred to as the origin, while the city that he finally reaches is referred to as the destination.

Q16. DIRECTIONS for questions 16 to 20: Select the correct alternative from the given choices.

If B must be the third city that Kiran visits (counting the origin as the first city), which of the following can be his destination?

- ☒ a) **G** ✓ Your answer is correct
- ☐ b) **E**
- ☐ c) **F**
- ☐ d) **A**

Q17. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
Which of the following pairs of cities cannot be the last two cities that he visits?

☐ a) **C, D**

☐ b) **E, F**

☐ c) **D, F**

☒ d) **More than one of the above** ✓ **Your answer is correct**

Q18. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.
If the origin is E, how many of the other seven cities can be the destination?

- ☐ a) 0
- ☐ b) 1
- ☒ c) 2 ✓ **Your answer is correct**
- ☐ d) 3

Q19. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

If the origin and the destination should not be directly connected, how many of the eight cities can be the origin?

☐ a) 4

☒ b) 5 **✗ Your answer is incorrect**

☐ c) 3

☐ d) 6

Q20. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

Kiran started from a city and his destination was not connected to the city from which he started. In how many ways can he visit all the cities?

☐ a) 10

☐ b) 11

☐ c) 9

☐ d) 12